

Ink & Motion

A Non-Photorealistic Rendering (NPR) system driven by physics-based dynamics and procedural stylization

1. System Overview

This system introduces temporal behavior into an otherwise static Non-Photorealistic Rendering (NPR) study by driving an ink-wash style figure with a physics-based dynamics loop rather than fixed keyframes. Three coupled subsystems run every frame: (1) a rigid-body-style physics integrator controlling position and velocity, (2) a procedural deformation layer producing volume-conserving squash and stretch, and (3) a particle system generating ink-splatter droplets on ground contact. A persistent, non-cleared canvas layer accumulates faint stains left by particles, giving the stylization a visible memory of the motion history.

2. Animation System

Physics-based motion

The figure's vertical motion is integrated with semi-implicit Euler stepping: gravity accelerates the point mass every frame, position updates from velocity, and each ground contact reflects the vertical velocity by a restitution coefficient e (0.4-0.95, adjustable). This produces a genuine bouncing-ball trajectory rather than an authored curve, so the same rules that make a real ball's flight path predictable govern this character.

Procedural deformation

Impact speed feeds a secondary damped spring-mass system that drives squash and stretch. On contact, the vertical scale is driven down proportionally to impact speed; a spring (stiffness k , damping d) pulls it back toward the rest shape over the following frames. Horizontal scale is set to the reciprocal square root of vertical scale, which approximately conserves the figure's silhouette area - the classic 'squash and stretch' principle, produced procedurally instead of hand-animated, paired naturally with 'slow-in / slow-out' as the spring settles.

Procedural stylization layer

The outline is redrawn every frame from a ring of sample points, each offset by a small sum of sine functions in time ('line boil'). This is the mechanism that makes hand-drawn and ink animation read as alive even on a held pose. Shading is quantized into a flat base tone plus a single darker crescent with cross-hatching, rather than a smooth gradient, mimicking sumi-e brush shading conventions.

3. Particle & Deformation System

Each ground contact spawns a burst of ink-droplet particles sampled from an upward cone, sized and counted in proportion to impact speed. Every particle is itself integrated under gravity and linear drag and fades in alpha over its lifetime before either dissipating or striking the ground, where it deposits a soft, irregular stain on the

persistent trail layer. Over a run, this produces an accumulating record of where and how hard the figure has landed - a stylized but physically motivated byproduct of the dynamics, not a decorative overlay.

4. Motion Analysis

Because energy can only leave the system at a ground contact (scaled by e^2) and is never added, the system's total mechanical energy (kinetic + potential) should trace a decaying envelope over time: each bounce apex should be lower than the previous one by a roughly constant ratio. This ratio is tracked live and compared against the theoretical value e^2 derived from the restitution setting.

Metric	What it captures
Peak-to-peak energy ratio	Ratio of successive bounce apex heights; expected to converge to e^2
Scale trace (scaleX x scaleY)	Confirms the squash/stretch deformation stays approximately area-conserving
Time since last impact	Used to correlate deformation recovery with the spring's settling time
Particle count	Confirms particle lifetime and drag are removing particles rather than accumulating unbounded state

Across the sliders' ranges, raising restitution slows the energy decay and produces visibly more bounces before rest; raising the squash intensity increases peak deformation without changing the trajectory, confirming that the stylization layer is decoupled from - but driven by - the underlying physics.

5. Stability & Realism Evaluation

A run is classified as stable when three conditions hold simultaneously: successive bounce-energy peaks decrease monotonically, the peak-to-peak ratio stays close to the theoretical e^2 within tolerance, and vertical position converges to the ground line rather than oscillating indefinitely or diverging. All three are computed live rather than assumed:

- Monotonic decay - each apex height in the recorded sequence is compared to the previous one; a rise instead of a fall flags an energy-injection bug in the integrator.
- Ratio consistency - large deviation from e^2 indicates the discrete time-step is under-resolving fast impacts (a numerical-stability concern common to explicit integrators, not a stylization issue).
- Convergence to rest - once impact speed and rest-height error both fall under a small threshold, velocity is zeroed to avoid infinite-precision jitter at the resting state, which is the correct physical end-state for a damped, dissipative system.

Realism, separately from numerical stability, is evaluated against traditional animation principles rather than photorealistic accuracy: squash-and-stretch magnitude scales with impact force, deformation recovers with a spring-like ease rather than snapping instantly, and secondary motion (the particle splash and its residual stain) lags and decays after the primary motion, consistent with how weight and follow-through read in hand-drawn animation.

6. Summary

The deliverable demonstrates all three required elements for this milestone: a hybrid physics/procedural animation system, a physically motivated particle and deformation system, and live, quantitative motion and stability evaluation rather than a purely qualitative visual check. The NPR treatment (line boil, hatching, ink-trail accumulation) is intentionally decoupled from the physics layer, so the same dynamics core could drive a different stylization without modification.